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$$\begin{aligned} & \int x \sin^2(x) dx \\ &= \int x \cdot \frac{1 - \cos 2x}{2} dx \\ &= \frac{1}{2} \int x dx - \frac{1}{2} \int x \cos(2x) dx \\ & \quad \left\{ \begin{array}{l} u = x \\ dv = \cos(2x) dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = dx \\ v = \frac{\sin 2x}{2} \end{array} \right. \\ & \int x \cos(2x) dx = \frac{x}{2} \sin 2x - \frac{1}{2} \int \sin 2x dx \\ &= \frac{x}{2} \sin 2x + \frac{\cos 2x}{4} \\ & \int x \sin^2 x dx = \frac{x^2}{4} - \frac{1}{2} \left(\frac{x}{2} \sin 2x + \frac{1}{4} \cos 2x \right) + C \\ &= \frac{x^2}{4} - \frac{x}{4} \sin 2x - \frac{\cos 2x}{8} + C \end{aligned}$$

References